

SIT292 Module 5 Self-Assessment

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score	Review
Question 1	1/1	
Question 2	2/2	
Question 3	1/1	
Question 4	3/3	
Question 5	3/3	
Total	10/10	(100%)

Congratulations! You have achieved the minimum threshold for this module's self-assessment.

Use the "Print this results summary" and save your attempt as a pdf. You will need the printout showing all questions for your module submission.

Performance Summary

Exam Name:	SIT292 Module 5 Self-Assessment
Session ID:	1817048384574605
Exam Start:	Thu Aug 21 2025 13:55:14
Exam Stop:	Thu Aug 21 2025 14:58:47
Time Spent:	1:03:32

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Question 1

Let U be the set of vectors $(x^2, -y^2, z^2)$ with x, y, z in $\mathbb{R}$.

Determine whether or not U is a subspace of $\mathbb{R}^3$.

The set U :

- ☒ includes the zero vector
- ☒ is closed under vector addition
- ☐ is closed under multiplication
- ☐ is a **subspace** of $\mathbb{R}^3$
- ☒ is **not a subspace** of $\mathbb{R}^3$



Expected answer:

- ☒ includes the zero vector
- ☒ is closed under vector addition
- ☐ is closed under multiplication
- ☐ is a **subspace** of $\mathbb{R}^3$
- ☒ is **not a subspace** of $\mathbb{R}^3$

Score: 1/1 ✓

✓ Your answer is correct. You were awarded 1 mark.

You scored 1 mark for this part.

Question 2

Consider the vectors

$$\mathbf{x} = \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix}, \text{ and } \mathbf{y} = \begin{pmatrix} 0 \\ 4 \\ 1 \end{pmatrix}$$

Let $\mathbf{z} = \begin{pmatrix} -1 \\ 7 \\ -2 \end{pmatrix}$. Determine whether or not $\mathbf{z}$ lies in $U = \text{span}\{\mathbf{x}, \mathbf{y}\}$.

If $\mathbf{z}$ is in U , write it as a linear combination of $\mathbf{x}$ and $\mathbf{y}$. If it is not possible, enter 0 for both coefficients.

$$\mathbf{z} = \boxed{0} \text{ Expected answer: } \underline{\quad 0 \quad} \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix} + \boxed{0} \text{ Expected answer: } \underline{\quad 0 \quad} \begin{pmatrix} 0 \\ 4 \\ 1 \end{pmatrix}.$$

Score: 2/2 ✓

✓ You were awarded 1 mark.

✓ You were awarded 1 mark.

You scored 2 marks for this part.

Question 3

Determine whether the following subset of vectors is independent.

$$\left\{ \begin{pmatrix} -1 \\ 0 \\ 2 \\ -2 \end{pmatrix}, \begin{pmatrix} -3 \\ 1 \\ 4 \\ -7 \end{pmatrix}, \begin{pmatrix} -1 \\ -2 \\ 6 \\ 0 \end{pmatrix} \right\} \text{ in } \mathbb{R}^4$$

Provide your selection. Use the reveal answers button to check your reasoning.

- ☐ the subset of vectors is independent
- ☒ the vectors are linearly dependent



Expected answer:

- ☐ the subset of vectors is independent
- ☒ the vectors are linearly dependent

Score: 1/1 ✓

✓ You chose a correct answer. You were awarded **1** mark.

You scored **1** mark for this part.

Question 4

Find a basis and calculate the dimension for the following subspace of $\mathbb{R}^4$.

$$\text{span}\{(-2, -1, -2, -1), (-2, -3, 2, -5), (1, 2, -2, 5), (3, 0, 6, 3)\}$$

Enter your basis as rows of a matrix. Use the controls to increase the number of rows as necessary.

Rows: Columns:

$$\begin{pmatrix} \begin{matrix} 1 & 0 & 2 & 0 \end{matrix} \\ \begin{matrix} 0 & 1 & -2 & 0 \end{matrix} \\ \begin{matrix} 0 & 0 & 0 & 1 \end{matrix} \end{pmatrix}$$

Expected answer: $\begin{pmatrix} \begin{matrix} 1 & 0 & 2 & 1 \end{matrix} \\ \begin{matrix} 0 & 1 & -2 & 2 \end{matrix} \\ \begin{matrix} 0 & 0 & 0 & 1 \end{matrix} \end{pmatrix}$

What is the dimension of the subspace?

Expected answer: 3

Score: 3/3 ✓

✓ Correct. Your vectors are independent and span the entire subspace. You were awarded **3** marks.

You scored **3** marks for this part.

Question 5

Find (a, b, c, d) in $\mathbb{R}^4$ such that the given set is orthogonal.

$$\{ (2, 0, 2, -2), (1, -2, 1, 2), (3, 3, -1, 2), (a, b, c, d) \}$$

a =

Expected answer: -11/8

b =

Expected answer: 3/2

c =

Expected answer: 19/8

d =

Expected answer: 1

Score: 3/3 ✓

✓ Correct. The vector you've provided will make the set orthogonal. You were awarded **3** marks.

You scored **3** marks for this part.

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